

# malbo for the speedrun

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## 1 Idea

Optimize a lower bound on the cross entropy loss to improve learning.

### 1.1 Derivation

Generate an adapted sequence of probability vectors  $p_t$  and then observe token  $n_t$  and receive reward  $\log(p_{n_t})$ . This reward is bounded above by zero and has no lower bound, which is problematic since we want to optimize a lower bound. Instead derive an influence function by pretending to do an  $\epsilon$  mix-in of the uniform distribution over  $k$  tokens,  $p = \epsilon/k + (1 - \epsilon)\pi$ , with  $\pi > 0$ .

$$\begin{aligned}\log(p_{n_t}) &= \log\left(\frac{\epsilon}{k} + (1 - \epsilon)\pi_{n_t}\right) \\ &= \log\left(1 + k\frac{1 - \epsilon}{\epsilon}\pi_{n_t}\right) + \log\left(\frac{\epsilon}{k}\right).\end{aligned}$$

Shifting off  $\log(\epsilon/k)$  gives the lower and upper bounded reward  $r_t = \log(1 + c\pi_{n_t})$  where  $c = k(1 - \epsilon)/\epsilon$ . We can use this to make a betting process

$$K_T(v, \pi, b) = \prod_{t \leq T} 1 + b_t \left( \frac{\log(1 + c\pi_{n_t})}{v} - 1 \right) \quad (1)$$

which is a martingale when evaluated at  $v = \mathbb{E}[\log(1 + c\pi_n)]$  and any adapted betting sequence  $b \in [0, 1)^T$ . Note an independent betting process is run for each sequence in the batch and  $t$  indexes token positions.

Using a universal portfolio betting scheme (see appendix A) yields lower bound

$$\hat{v}(\theta) = \max_{\substack{v > 0 \\ b \in [0, 1]}} v \quad \text{s.t.} \quad \log K_T(v, \theta, b\vec{1}) \geq \log\left(\frac{1 + T}{\alpha}\right) \quad (2)$$

valid with probability at least  $(1 - \alpha)$ . As derived in appendix C,

$$\nabla_{\theta} \hat{v}(\theta) = \sum_{t \leq T} \left( \frac{\kappa_t}{\sum_{t' \leq T} \kappa_{t'}} \right) \gamma_t \nabla_{\theta} \log \pi(y_t | y_{< t}, x; \theta), \quad (3)$$

$$\kappa_t \doteq \frac{b^*(\hat{v}(\theta), \theta)}{\hat{v}(\theta) + b^*(\hat{v}(\theta), \theta) (r_t(\theta) - \hat{v}(\theta))} > 0, \quad (4)$$

$$\gamma_t \doteq \frac{c\pi(y_s | y_{< s}, x; \theta)}{1 + c\pi(y_s | y_{< s}, x; \theta)} \in [0, 1]. \quad (5)$$

This looks like a weighted version of the standard cross-entropy loss. Note  $\kappa_t \gamma_t$  is decreasing in  $\pi$  so this emphasizes rare tokens.

## A Universal Portfolio Strategy

Equation (1) corresponds to the returns of a portfolio over two assets: a  $(1 - b_t)$  allocation to cash, which returns 1; and a  $b_t$  allocation to a stock which returns  $v^{-1} \log(1 + c\pi(y_t | y_{< t}, x; \theta))$ . A universal portfolio (Cover 1991) betting strategy is given by

$$\hat{b}_t(v, \theta) = \frac{\int_0^1 z K_{t-1}(v, \theta, z \vec{1}) dz}{\int_0^1 K_{t-1}(v, \theta, z \vec{1}) dz},$$

where  $z$  is a scalar and we evaluate  $K_{t-1}(v, \theta, z \vec{1})$  at the fixed constant bet  $z$ . This satisfies the conditions of Theorem 10.3 of Cesa-Bianchi et al. (2006), therefore

$$\log K_T(v, \theta, \hat{b}(v, \theta)) \geq \log K_T(v, \theta, b^* \vec{1}) - \log(1 + T),$$

where  $b^*$  is any fixed bet in hindsight. Combining this Ville's inequality yields yields eq. (2).

## B Solving eq. (2)

### B.1 eq. (2) defines a convex feasibility region

Consider  $v_1 < v_2$ , then

$$\begin{aligned} \log K_t(v_2, \theta, b^*(v_2, \theta)) &\leq \log K_t(v_1, \theta, b^*(v_2, \theta)) && \text{(decreasing in } v) \\ &\leq \log K_t(v_1, \theta, b^*(v_1, \theta)) && \text{(optimality of } b^*) \end{aligned}$$

therefore the feasibility set is either  $\emptyset$  or  $(0, v^*]$  for some  $v^* > 0$ .

## B.2 eq. (2) defines a non-empty feasible region

Let  $r_s(\theta) \doteq \log(1 + c\pi(y_s|y_{<s}, x; \theta))$ . By assumption  $\pi > 0$  ergo  $r_s(\theta) > 0$ . Let  $s'$  denote the largest  $r_s(\theta)$ . For  $Tv < r_{s'}(\theta)$ , lower bound the martingale wealth by

$$\begin{aligned} \log K_t(v, \theta, b\vec{1}) &\geq \log \left( 1 + b \left( \frac{r_{s'}(\theta)}{v} - 1 \right) \right) + (T-1) \log(1-b) \\ &\geq \log \left( \frac{r_{s'}(\theta)}{Tv} \right) + (T-1) \log \left( \frac{(T-1)r_{s'}(\theta)}{T(r_{s'}(\theta) - v)} \right) \quad (\dagger) \\ &\geq \log \left( \frac{r_{s'}(\theta)}{Tv} \right) + (T-1) \log \left( \frac{T-1}{T} \right) \\ &\geq \log \left( \frac{r_{s'}(\theta)}{Tv} e^{-1} \right) \end{aligned}$$

where  $(\dagger)$  follows from choosing  $b = \frac{r_s(\theta) - Tv}{T(r_s(\theta) - v)} \in [0, \frac{1}{T}]$ . Equating this bound with the threshold yields a lower bound

$$\hat{v}(\theta) \geq \frac{\alpha e^{-1}}{T(1+T)} r_{s'}(\theta)$$

with corresponding bet  $b > (T+1)^{-1}$ .

## B.3 Empirical mean is not a lower bound

Let  $r_s(\theta) \doteq \log(1 + c\pi(y_s|y_{<s}, x; \theta))$ . By assumption  $\pi > 0$  ergo  $r_s(\theta) > 0$ . From Jensen's inequality, for a constant bet  $b$ ,

$$\begin{aligned} \frac{1}{t} \log K_t(v, \theta, b\vec{1}) &= \frac{1}{t} \sum_{s \leq t} \log \left( 1 + b \left( \frac{r_s(\theta)}{v} - 1 \right) \right) \\ &\leq \log \left( 1 + \frac{1}{t} \sum_{s \leq t} b \left( \frac{r_s(\theta)}{v} - 1 \right) \right) \\ &= \log \left( 1 - b + b \frac{t^{-1} \sum_{s \leq t} r_s(\theta)}{v} \right) \\ &= 0, \quad (\dagger) \end{aligned}$$

where  $(\dagger)$  is from evaluating at  $v = t^{-1} \sum_{s \leq t} r_s(\theta)$ . This is standard result in portfolio theory about the optimality of an all cash portfolio, see section 15.2 of Cover (1999).

## B.4 Computing $\hat{v}(\theta)$

Let  $r_s(\theta) \doteq \log(1 + c\pi(y_s|y_{<s}, x; \theta))$ . Because  $(\alpha e^{-1}/(T(1+T))) \max_s r_s(\theta) < \hat{v}(\theta) < t^{-1} \sum_{s \leq t} r_s(\theta)$ , we can efficiently compute  $\hat{v}(\theta)$  via Brent's method (Brent

1973), root-finding on the function

$$g(v) \doteq \log \left( \frac{\alpha}{1+T} \right) + \max_{b \in [0,1]} \log K_T(v, \theta, b\vec{1}).$$

The inner maximization can also be done via Brent's method, root-finding on the gradient

$$\frac{\partial}{\partial b} \log K_T(v, \theta, b\vec{1}) = \frac{1}{v} \sum_{s \leq T} \frac{r_s(\theta) - 1}{1 + b \left( \frac{r_s(\theta)}{v} - 1 \right)}.$$

On a GPU, vectorization is more important than algorithmic constants, so using a fixed number of (nested) bisections instead of Brent's method is preferred.

## C Differentiating eq. (2)

The constraint set for eq. (2) is convex (see appendix B.1) and not empty (see appendix B.2). Clearly  $b = 0$  is infeasible due to the wealth constraint. The remaining Lagrangian is

$$\mathcal{L}(v, b) = v - \lambda g(v, b) - \mu(b - 1),$$

where  $g(v, b) = \log((1+T)/\alpha) - \log K_T(v, \theta, b\vec{1})$  is the wealth constraint.  $v$ -stationarity implies

$$1 = -\lambda^* \left. \frac{\partial}{\partial v} \log K_T(v, \theta, b\vec{1}) \right|_{v=v^*, b=b^*}.$$

Via the Envelope Theorem (Mas-Colell et al. 1995),

$$\nabla_{\theta} \hat{v}(\theta) = \lambda^* \left. \nabla_{\theta} \log K_T(v, \theta, b\vec{1}) \right|_{\substack{v=\hat{v}(\theta) \\ b=b^*(\hat{v}(\theta), \theta)}},$$

Let  $r_s(\theta) \doteq \log(1 + c\pi(y_s|y_{<s}, x; \theta)) > 0$ . The terms are

$$\begin{aligned} \frac{\partial \log K_T(v, \theta, b\vec{1})}{\partial v} &= -\frac{1}{v} \sum_{s \leq T} \frac{br_s(\theta)}{v + b(r_s(\theta) - v)}, \\ \nabla_{\theta} \log K_T(v, \theta, b\vec{1}) &= \sum_{s \leq T} \frac{b}{v + b(r_s(\theta) - v)} \nabla_{\theta} r_s(\theta). \end{aligned}$$

which yields

$$\begin{aligned} \nabla_{\theta} \hat{v}(\theta) &= \hat{v}(\theta) \sum_{s \leq T} \frac{\kappa_s}{\sum_{s' \leq T} r_{s'} \kappa_{s'}} \nabla_{\theta} r_s(\theta) \\ \kappa_t &\doteq \frac{b^*(\hat{v}(\theta), \theta)}{\hat{v}(\theta) + b^*(\hat{v}(\theta), \theta) (r_t(\theta) - \hat{v}(\theta))} > 0. \end{aligned}$$

Since all rewards are non-zero,  $b^* \in (0, 1)$ , so  $b$ -stationarity implies

$$\sum_{s \leq T} \frac{r_s/v^* - 1}{1 + b^*(r_s/v^* - 1)} = 0$$

$$\sum_{s \leq T} r_s \kappa_s = v^* \sum_{s \leq T} \kappa_s$$

Therefore

$$\nabla_{\theta} \hat{v}(\theta) = \sum_{s \leq T} \frac{\kappa_s}{\sum_{s' \leq T} \kappa_{s'}} \nabla_{\theta} r_s(\theta).$$

Finally note

$$\nabla_{\theta} r_s(\theta) = \gamma_s \nabla_{\theta} \log \pi(y_s | y_{<s}, x; \theta),$$

$$\gamma_s \doteq \frac{c\pi(y_s | y_{<s}, x; \theta)}{1 + c\pi(y_s | y_{<s}, x; \theta)} \in [0, 1].$$